

Name _____ Date _____ Period _____

Graphing $f(x) = |x|$ and Key Attributes

Algebra II | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from a table and identify its domain, range, intercepts, symmetry, and minimum value.

Vocabulary

Absolute value — the distance a number is from _____, always _____ (≥ 0).

Vertex — the point where the two straight-line pieces of $f(x) = |x|$ meet: _____.

Key attributes — domain, range, _____, symmetry, and minimum/maximum.

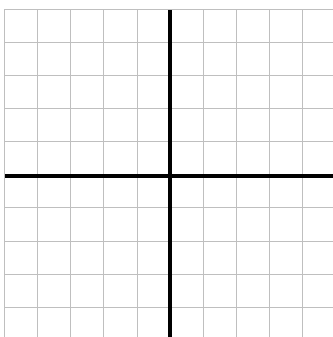
Symmetric about the y-axis — the left and right halves of the graph are _____.

Part 1 — Build the Table, Then Graph

Fill in $f(x) = |x|$ for each x -value.

x	-3	-2	-1	0	1	2	3
$f(x)$							

Plot the seven points and connect them. Each square is 1 unit; both axes run -5 to 5.



Part 2 — Key Attributes of $f(x) = |x|$

Domain: _____

Range: _____

x-intercept: _____ y-intercept: _____

Vertex: _____ Symmetric about: _____

Minimum value: _____ Maximum value: _____

Part 3 — What Does $|x|$ Mean in Real Life? Distance

A hiking trail runs both directions from a trailhead at position 0. A hiker's position is x miles (negative = one direction, positive = the other). The distance from the trailhead is $d(x) = |x|$.

Hiker at $x = -3.5$: distance = _____

Hiker at $x = 6$: distance = _____

A thermometer reads t degrees relative to 0°C (freezing). $|t|$ tells you how many degrees you are from freezing, above or below.

$t = -15$: _____ degrees from freezing $t = 8$: _____ degrees from freezing

In your own words, explain why distance can never come out negative.

Sample: _____

Part 4 — Compare to a Line

Fill in both columns to see what changes and what stays the same.

	$y = x$ (a line)	$f(x) = x $
Domain	_____	_____
Range	_____	_____
x- and y-intercept	_____	_____
Minimum?	_____	_____
Symmetric?	_____	_____

Practice

- $|-8| =$ _____
- $|6| =$ _____
- $|0| =$ _____
- $|-2.5| =$ _____
- If $f(x) = |x|$ and $x = -10$, $f(x) =$ _____
- $f(x) = |x|$ has a vertex at $(0, 0)$. Give one more point on each side of it. Left: _____ Right: _____
- True or False: the graph of $f(x) = |x|$ is symmetric about the x-axis. Explain.
False _____
- The points $(-4, 4)$ and $(4, 4)$ are both on the graph of $f(x) = |x|$. What key attribute does this illustrate?
Symmetry about the y-axis _____
- One point is at $x = -6$, a certain distance from zero. What x-value on the OTHER side of zero is the same distance away? _____
- Which is larger, $|-15|$ or $|9|$? _____
- Put these in order from least to greatest: $|-4|$, $|1|$, $|-9|$, $|6|$

Exit Ticket

For $f(x) = |x|$, give the domain, range, vertex, and axis of symmetry.

Domain _____ Range _____

Vertex _____ Axis of symmetry _____

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Transformations of $f(x) = |x|$, Part I

Algebra II | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe and graph the effect of a and b on $f(x) = a|bx|$ - vertical/horizontal stretches, compressions, and reflections.

Vocabulary

Transformation — a change to a parent function's graph: _____.

Vertical stretch/compression — $af(x)$; $|a| > 1$ _____ away from the x-axis, $0 < |a| < 1$ _____ toward it.

Horizontal stretch/compression — $f(bx)$; $|b| > 1$ _____ toward the y-axis, $0 < |b| < 1$ _____ away from it.

Reflection — $a < 0$ flips over the _____; $b < 0$ flips over the _____.

Part 1 — The Effect of a in $f(x) = a|x|$

Function	a-value	Effect	Vertex
$f(x) = x $	1	parent function	(0, 0)
$f(x) = 2 x $	2	_____	_____
$f(x) = (1/2) x $	1/2	_____	_____
$f(x) = - x $	-1	_____	_____

Notice: no matter what a is, the vertex never moves - $a(0)$ is always 0.

You Try. Describe $f(x) = 4|x|$: _____ Describe $f(x) = -(1/3)|x|$: _____

Part 2 — The Effect of b in $f(x) = |bx|$

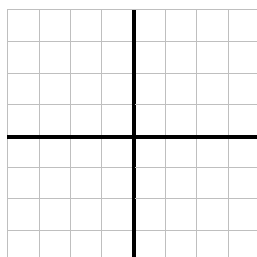
Function	b-value	Effect
$f(x) = x $	1	parent function
$f(x) = 2x $	2	_____
$f(x) = (1/2)x $	1/2	_____
$f(x) = -x $	-1	_____

You Try. Describe $f(x) = |3x|$: _____ Describe $f(x) = |(1/5)x|$: _____

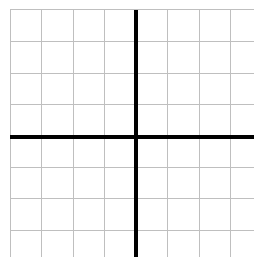
Part 3 — Graph and Compare

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5. Which one looks steeper?

A. $f(x) = 2|x|$



B. $f(x) = (1/2)|x|$



Steeper graph: _____

Part 4 — Rank Them

Order from FLATTEST to STEEPEST: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$

Practice

1. Describe the transformation of $f(x) = 5|x|$ from the parent. _____
2. Describe the transformation of $f(x) = |(1/4)x|$ from the parent. _____
3. Describe the transformation of $f(x) = -6|x|$ from the parent.

4. Which of these two graphs is steeper: $f(x) = 3|x|$ or $f(x) = (1/5)|x|$?

5. $f(x) = -4|2x|$ combines an a-effect and a b-effect. Describe both.

6. True or False: $f(x) = |-5x|$ looks different from $f(x) = 5|x|$. Explain.

7. Describe the transformation of $f(x) = (1/3)|-x|$ from the parent.

8. Match each function to its effect: $f(x) = 7|x|$, $f(x) = |7x|$, $f(x) = -|x|$, $f(x) = (1/7)|x|$.
A. vertical compression B. reflection only C. vertical stretch by 7 D. horizontal compression
 $f(x) = 7|x|$: ____ $f(x) = |7x|$: ____ $f(x) = -|x|$: ____ $f(x) = (1/7)|x|$: ____
9. Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look IDENTICAL for every x you try. What must be true of a and b ?

10. Describe the transformation of $f(x) = -(2/3)|x|$, using both the stretch/compression amount and the reflection.

Exit Ticket

Describe every transformation in $f(x) = -3|x|$ from the parent function (stretch/compression amount, and any reflection).

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Transformations of $f(x) = |x|$, Part II

Algebra II | Unit 2 | Day 3 | TEK A2.6.C

Objective: I can describe and graph the effect of c and d on $f(x) = a|x-c|+d$, and combine all four transformations into one function.

Vocabulary

Horizontal shift — $f(x-c)$ moves the graph left/right. $c>0$ shifts _____, $c<0$ shifts _____ - opposite the sign.

Vertical shift — $f(x)+d$ moves the graph up/down. $d>0$ shifts _____, $d<0$ shifts _____ - same as the sign.

Vertex, general form — for $f(x) = a|x-c|+d$, the vertex is always at _____.

Part 1 — Horizontal Shift: $f(x - c)$

Function	Shift	Vertex
$f(x) = x - 3 $	_____	_____
$f(x) = x + 2 $	_____	_____
$f(x) = x - 7 $	_____	_____

Part 2 — Vertical Shift: $f(x) + d$

Function	Shift	Vertex
$f(x) = x + 4$	_____	_____
$f(x) = x - 5$	_____	_____
$f(x) = x + 1.5$	_____	_____

Part 3 — Combine All Four: $f(x) = a|x - c| + d$

"Compressed by $1/3$, shifted down 5, shifted left 2" becomes one function:

$f(x) =$ _____ vertex: _____

You Try. Describe $f(x) = -2|x - 1| + 6$, and give its vertex.

_____ vertex: _____

"Stretched by 4, shifted up 2, shifted right 5" becomes one function:

$f(x) =$ _____ vertex: _____

Part 4 — Real World: Dunhill Drafting's Roofline

Roofline data was modeled by trial-and-error transformation, landing on $f(x) = -1.2|x - 3| + 8$.

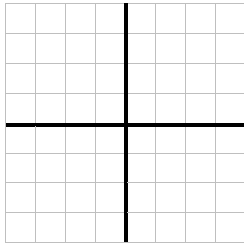
Vertex (highest point of the roof): _____

Describe the transformations from the parent:

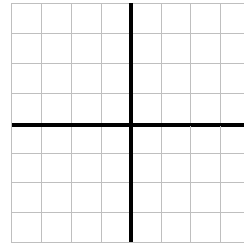
Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = |x - 2| - 3$



B. $f(x) = |x + 3| + 2$



Vertex of A: _____ Vertex of B: _____

Practice

1. Give the vertex of $f(x) = |x + 5| - 1$. _____
2. Give the vertex of $f(x) = 4|x - 6| + 2$. _____
3. Write a function with vertex $(-4, 7)$ and no stretch/reflection. _____
4. Give the vertex of $f(x) = -3|x + 8| - 6$. _____
5. Write a function with vertex $(2, -9)$ and a vertical stretch by 5. _____
6. Give the vertex of $f(x) = (1/4)|x - 10| + 3$. _____
7. A function has vertex $(-6, 0)$ and opens downward with no stretch. Write it. _____
8. True or False: every function of the form $a|x - c| + d$ has the SAME shape as the parent, just moved and resized. Explain. _____
9. Give the vertex of $f(x) = 2|x + 1| - 4$, and state whether it opens up or down. _____
10. Write a function with vertex $(0, -3)$ that is a reflection of the parent with no stretch.

Exit Ticket

Describe every transformation of $f(x) = -(1/2)|x + 3| - 4$ from the parent, and give its vertex.

_____ vertex _____

Formulating Absolute Value Linear Equations

Algebra II | Unit 2 | Day 4 | TEK A2.6.D

Objective: I can formulate (set up, without solving) an absolute value linear equation from a real-world description.

Vocabulary

Formulate — set up an equation from a description - _____.**Center value** — the target/ideal number the situation is measured against.**Margin** — how far off from the center is still acceptable.**Template** — every problem in this lesson fits the pattern $|x - \text{center}| = \text{margin}$.

Part 1 — Clarissa's New House

Clarissa wants a house within 5 houses of house #1248 on an even-numbered street.

Center value: _____ Margin: _____

Equation: _____

Part 2 — Surveying Margin of Error

*A surveying instrument measures a 10-mile stretch with a 1.5% margin of error.*The 1.5% margin of error is measured as a percent of the true 10-mile distance x , so it is written $1.5 = |x - 10|/10 \cdot 100$.

Equation: _____

Part 3 — Brighton Builders' Hallway

A hallway has offices on both sides, evenly spaced from the center at $15x$ feet, and the total layout must equal 292 feet with 52 feet reserved for walls and doors.

Equation: _____

Practice — Formulate, Do Not Solve

1. A recipe calls for 2 cups of flour, with acceptable measurement error of 0.25 cups.

Center: _____ Margin: _____ Equation: _____

2. A shipping company's box-weight target is 50 lb, with a tolerance of 3 lb.

Equation: _____

3. A factory's bolt length must be within 0.02 cm of 6 cm.

Equation: _____

Exit Ticket

A thermostat targets 70°F, with an acceptable range of 2°F above or below. Formulate an equation - do not solve.

Center: _____ Margin: _____

Equation: _____

Solving Absolute Value Linear Equations

Algebra II | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can solve absolute value linear equations by isolating the absolute value and splitting into two cases.

Vocabulary

Isolate — get the absolute value expression _____ on one side before splitting.**Split into two cases** — $|A| = k$ ($k > 0$) becomes $A = k$ _____ $A =$ _____.**Check** — plug each solution back into the _____ equation before accepting it.

Part 1 — The Method

*Step 1: Isolate the absolute value. Step 2: Split into two cases. Step 3: Solve each. Step 4: Check.***Example 1.** $|2x - 3| - 4 = 3$ Isolate: $|2x - 3| =$ _____Split: $2x - 3 = 7$ OR $2x - 3 =$ _____Solve: $x =$ _____ OR $x =$ _____ Solution set: { _____ }

Part 2 — More Examples

Example 2. $|x| + 2 = 5$ Isolate: $|x| =$ _____ Split/Solve: $x =$ _____ or $x =$ _____ (both valid)**You Try.** $|3x + 6| = 12$ $3x + 6 = 12$ or $3x + 6 =$ _____ $x =$ _____ or $x =$ _____

Part 3 — Isolate First, Then Split

Example 3. $2|x + 1| - 3 = 7$ Isolate: $2|x + 1| =$ _____ $\rightarrow |x + 1| =$ _____Split/Solve: $x + 1 = 5$ or $x + 1 =$ _____ $x =$ _____ or $x =$ _____**You Try.** $3|x - 4| + 1 = 13$ $|x - 4| =$ _____ $x =$ _____ or $x =$ _____

Practice

1. $|x - 4| = 9$ $x =$ _____ or $x =$ _____

2. $|5x| - 2 = 18$ $x =$ _____ or $x =$ _____

3. $4|x + 2| = 20$ $x =$ _____ or $x =$ _____

Exit Ticket

Solve $2|x - 5| - 1 = 9$, showing the isolate and split steps.

$|x - 5| =$ _____ $x =$ _____ or $x =$ _____

Extraneous Solutions and Real-World Applications

Algebra II | Unit 2 | Day 6 | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions when solving absolute value equations, and solve the real-world equations from Day 4.**Vocabulary****Extraneous solution** — a value the algebra produces that does _____ actually satisfy the _____ equation.**Why it happens here** — splitting into two cases can create a solution where the "inside" expression has the _____ for that case.**Part 1 — A Solution That Does Not Check Out**Solve $3|x - 1| - 2 = 6x + 1$ and check BOTH answers in the original equation.Isolate: $3|x - 1| = \underline{\hspace{2cm}}$ $\rightarrow |x - 1| = 2x + 1$ Split: $x - 1 = 2x + 1$ or $x - 1 = \underline{\hspace{2cm}}$ Solve: $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Check $x = -2$: $3|-3| - 2 = 7$, $6(-2) + 1 = -11$. $7 \neq -11 \rightarrow \underline{\hspace{2cm}}$ Check $x = 0$: $3|-1| - 2 = 1$, $6(0) + 1 = 1$. $1 = 1 \rightarrow \underline{\hspace{2cm}}$ Solution set: $\{\underline{\hspace{2cm}}\}$ **Part 2 — Solve Day 4's Real-World Equations**Clarissa's houses: $|x - 1248| = 5$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Both are even-numbered addresses within 5 houses, so
 $\underline{\hspace{2cm}}$ **Part 3 — Surveying and Brighton Builders**Surveying: $1.5 = |x - 10|/10 \cdot 100$ $|x - 10| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ milesBrighton Builders: $2|15x| + 52 = 292$ $|15x| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ (x is a suite count, so use $x = 8$)That is 8 office suites per side per floor $\rightarrow \underline{\hspace{2cm}}$ suites per floor $\rightarrow \underline{\hspace{2cm}}$ total suites in the 20-story building**Practice**1. Solve $|x + 3| = 2x$ and check both solutions. $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Solution set: $\{\underline{\hspace{2cm}}\}$ 2. Solve $|2x - 1| = x + 4$ and check both solutions. $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Solution set: $\{\underline{\hspace{2cm}}\}$ (both valid this time)**Exit Ticket**Solve $|x - 4| = 3x$, checking both solutions in the ORIGINAL equation. State the solution set. $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ Solution set: $\{\underline{\hspace{2cm}}\}$

Solving Absolute Value Linear Inequalities

Algebra II | Unit 2 | Day 7 | TEK A2.6.F

Objective: I can solve absolute value linear inequalities using the graphing method and the algebraic (compound inequality) method.

Vocabulary

Compound inequality — two inequalities joined by _____ or _____.

"Or" statement — true if EITHER part is true - used for _____ or _____.

"And" statement — true only if BOTH parts are true - used for _____ or _____.

Part 1 — The Graphing Method

Compare $y1 = |expression|$ to $y2 = constant$. Where $y1$ is BELOW $y2$, that is the "less than" solution; where $y1$ is ABOVE $y2$, that is the "greater than" solution.

Part 2 — The Algebraic Method

Example 1 ($<$, uses AND). $|2x + 3| < 5$

$$-5 < 2x + 3 < \underline{\hspace{2cm}} \rightarrow -8 < 2x < 2 \rightarrow \underline{\hspace{2cm}}$$

Example 2 ($\geq$, uses OR). $|2x - 3| + 4 \geq 9$

Isolate: $|2x - 3| \geq \underline{\hspace{2cm}}$

$$2x - 3 \leq -5 \text{ or } 2x - 3 \geq \underline{\hspace{2cm}} \rightarrow \underline{\hspace{2cm}}$$

Part 3 — You Try

1. $|x + 4| \leq 6$ _____

2. $|3x - 1| > 8$ _____

3. $|x - 5| + 2 \geq 9$ _____

Practice

1. $|2x| < 10$ _____

2. $|x + 1| \geq 3$ _____

3. $3|x - 2| < 12$ _____

4. Is $x = 0$ a solution to $|x - 3| > 5$? _____

Exit Ticket

Solve $|3x + 2| \geq 7$, and state whether it is an "and" or "or" compound inequality.

Absolute Value Inequalities: Special Cases and Interval Notation

Algebra II | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, solve harder cases, and recognize when an inequality has no solution.

Vocabulary

No solution — NO value of x works. Happens when an isolated absolute value is set ____ or ____ a negative number.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$ x > 5$	_____	_____
$ x \geq 5$	_____	_____
$ x < 5$	_____	_____
$ x \leq 5$	_____	_____

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \leq 2x - 1$

Isolate: $|3x + 1| \leq$ _____

$-(2x + 3) \leq 3x + 1 \leq 2x + 3$

Solve both halves and combine: _____

Part 3 — When There Is No Solution

Example. $4 + |x + 1| \leq 2$

Isolate: $|x + 1| \leq$ _____

An absolute value can never be negative, so it can never be $\leq$ a negative number. Solution:

Practice

- Write $|x| \geq 8$ in interval notation. _____
- Write $|x| < 3$ in interval notation. _____
- Solve $|x - 2| + 6 \leq 1$ _____
- Solve $|2x - 1| > x + 5$ _____

Exit Ticket

Solve $3 + |x - 4| < 1$, and explain why the answer comes out the way it does.
